

Discuss fixed ratio blood product transfusion versus goal directed factor concentrate administration in the management of a severely injured polytrauma patient.

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Sample Answer

In the management of a severely injured polytrauma patient,

Fixed ratio blood product transfusion

- Refers to the empirical transfusion of FFP, Platelets, RBCs in 1:1:1 ratio
- Aims at early correction (with FFP and platelets) of trauma-induced coagulopathy
- Typically implemented as part of hospital's Massive Transfusion Protocol (MTP)
- Standard of care for Damage Control Resuscitation (DCR) in trauma patients.

However, there are several issues surrounding this approach:

- Inadequate resuscitation in terms of factor replacement, especially fibrinogen, which is depleted earlier/ more in trauma-induced coagulopathy.
- Over-resuscitation in terms of volume load, leading to worsening haemorrhage.
- Delay in FFP administration (compared to RBC, platelets) due to thawing time
- Inconclusive evidence on ideal component ratio for transfusion (PROMMTT, PROPPR trial)
- Complications associated with blood product transfusion
 - Transfusion-associated circulatory overload (TACO)
 - Transfusion-associated acute lung injury (TRALI)
 - Transmission of infectious diseases

As such, there is a trend towards using goal directed factor concentrate-based management:

- (i) **Goal-directed** approach (instead of empirical administration)
 - a. Using point-of-care visco-elastic measurement such as ROTEM, TEG with quick TAT
 - b. Allows for more precise replacement of deficient components
 - c. Reduces wastage of blood products
- (ii) **Factor concentrate-based** management (instead of FFP)
 - a. Prothrombin complex concentrate (PCC) and Fibrinogen Concentrate (FC) can be made readily available in OT, avoiding need for long thawing time for FFP.
 - b. Smaller administration volume, which facilitates permissive hypotension and minimises haemodilution
 - c. No need for cross-matching. Avoids complications associated with blood product transfusion.

Given its potential benefits, medical institutions with established ROTEM/TEG-based treatment algorithm, specialised equipment/drugs, and trained personnel should adopt a **goal directed factor concentrate-based management**. Fixed ratio blood product transfusion may still be useful during initial haemostatic resuscitation or if visco-elastic assays are normal.